

Aman Sood

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EDUCATION

Punjab Engineering College (PEC)

B.E. in Electrical Engineering

Chandigarh, India

Aug 2023 – Jun 2027

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Probability & Statistics, Linear Algebra

Class 12 (CBSE): 85% | Class 10 (ICSE): 95%

WORK EXPERIENCE

Bharat Electronics Limited (BEL)

Engineering Intern

India

Jan 2026 – May 2026

- Analyzed operational datasets across substations, DG sets, and solar systems to surface performance anomalies and deliver actionable insights for stakeholders.
- Built structured energy-data tracking and documentation workflows supporting ISO 50001 compliance readiness and measurable sustainability outcomes.
- Translated multi-source sensor and operational data into review-ready summaries; contributed to work recognized at the GreenTech Energy Awards.

PROJECTS

E32Log — Real-Time Vehicle Telemetry & OBD-II Logger

[GitHub](#)

C++, Python, ESP32, OBD-II

- Built a real-time telemetry pipeline ingesting 10+ vehicle parameters (RPM, speed, coolant temp, fuel efficiency) over OBD-II on ESP32 with sub-second polling.
- Logged and streamed 1000+ live diagnostic records via wireless/local output for continuous monitoring and debugging.
- Designed modular OOP classes for acquisition, buffering, and output to keep end-to-end latency low on constrained hardware.

Intelligent Traffic Surveillance System

Python, YOLOv8, DeepSORT, OpenCV, Edge AI

- Built a real-time multi-vehicle detection and tracking pipeline with YOLOv8 + DeepSORT for traffic monitoring and rule-based violation analytics.
- Implemented ROI/polygon counting plus ambulance-aware exception logic to cut false positives in automated red-light enforcement at the edge.

E-Commerce Web Application

[Live](#) | [GitHub](#)

JavaScript, Node.js, Express, REST APIs

- Shipped a responsive full-stack storefront with catalog, cart, and checkout flows; integrated REST APIs with clean request/response handling.
- Added server-side cart validation (shipping/tax), order-creation endpoints, and a light/dark theme toggle for a production-ready UX.

Personal Portfolio Website

[Live](#) | [GitHub](#)

Next.js, TypeScript, Framer Motion

- Built a responsive portfolio with animated about/skills/projects/contact sections, light/dark theme, and one-click resume download.

Skills & Certifications

Languages: Python, C++, SQL, Java, JavaScript, TypeScript

AI / Data: Pandas, NumPy, Scikit-learn, Statistical Analysis, OpenCV, YOLOv8, Prompt Engineering, RAG fundamentals

Engineering: DSA & OOP, FastAPI, Node.js, REST APIs, PostgreSQL, MongoDB, Next.js, React, Git, Linux, Docker, ESP32 / IoT

Interests: AI systems, edge computing, data products, shipping end-to-end prototypes